

Assignment 4 Writeup

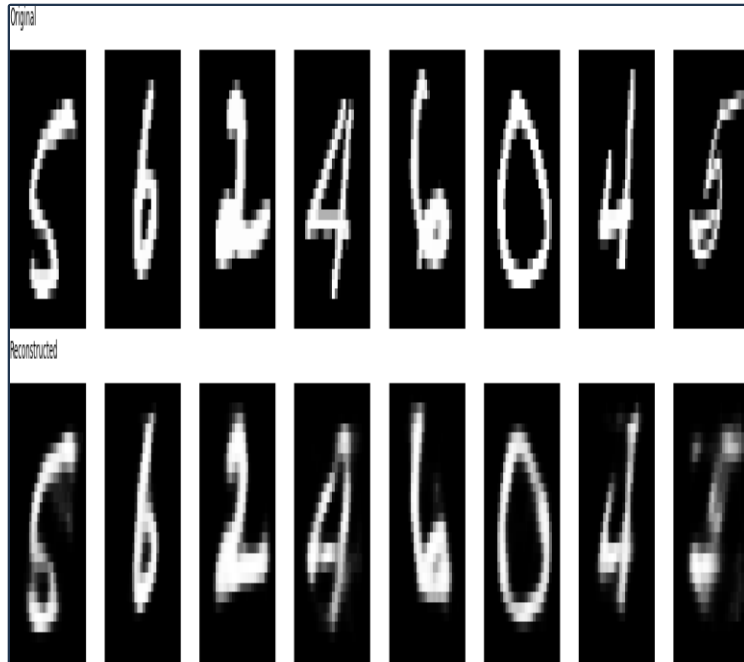
DO NOT TAG

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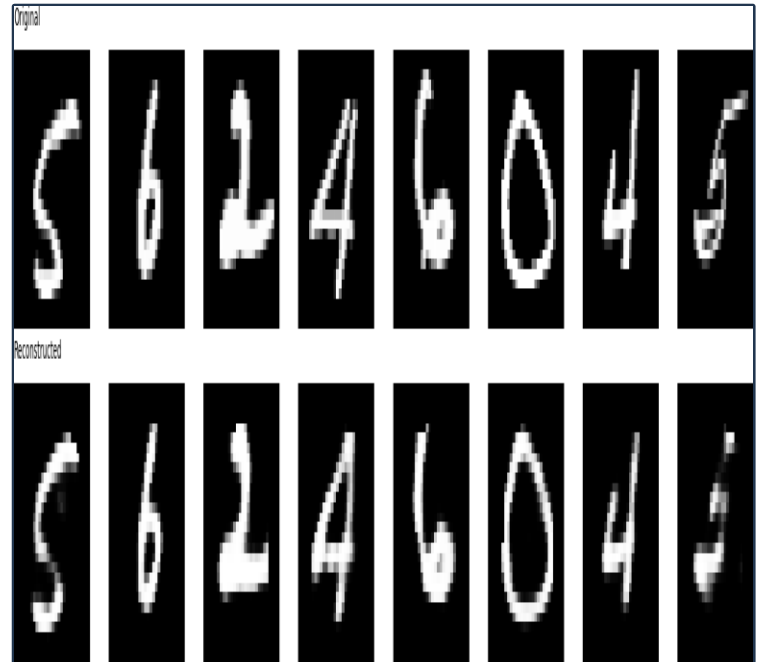
Variational Autoencoder (VAE) Q1 - MNIST

1. Tune beta value with L2 reconstruction loss.
2. Keeping beta fixed switch reconstruction loss to L1.
3. Show examples of L1 and L2 recon loss:

L2 Recon Loss



L1 Recon Loss



Variational Autoencoder (VAE) Q1 cont.

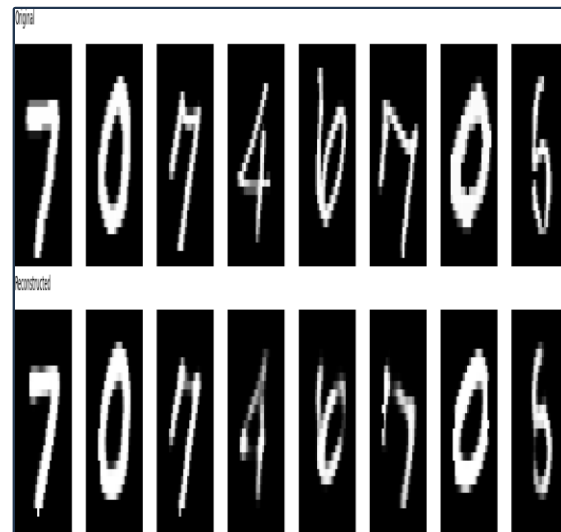
When comparing the performance of the VAE using L1 (MAE) and L2 (MSE) reconstruction losses, several differences were observed. L1 loss produced sharper and more defined reconstructions, whereas L2 loss resulted in smoother images but introduced noticeable blurring, especially around the edges of the digits. L1 loss proved to be more robust to outliers by penalizing large errors linearly, preventing excessive smoothing, whereas L2 loss, which penalizes large errors quadratically, was more sensitive to outliers, leading to less sharp reconstructions. Visual analysis confirmed that the L1-reconstructed images preserved finer details better than those generated using L2, where the overall structure was maintained but at the cost of clarity. Additionally, switching to L1 loss led to smoother and more stable convergence during training. While L2 loss showed higher initial KL divergence, it gradually decreased but still resulted in slightly inferior image quality compared to L1.

Variational Autoencoder (VAE) Q2 - MNIST

As we optimize for our few-shot setting (KNN classifier) what happens to the quality of generated samples? Share an example and explain your observations.

Answer:

In the few-shot setting optimized for the KNN classifier, the VAE's reconstructions remain largely similar to the original digits, though some subtle blurring and deformation appear. For example, the digit "7" retains its overall shape and stroke direction, while the "0" stays neatly enclosed and consistent with the input image. However, digits like "4" and "6" exhibit mild distortions and less crisp edges compared to the originals, suggesting that limited data slightly impacts the model's ability to capture finer details. Despite this minor difference, the latent space still encodes enough distinguishing features for high KNN accuracy, indicating that the VAE can learn meaningful representations even with only ten samples per class.

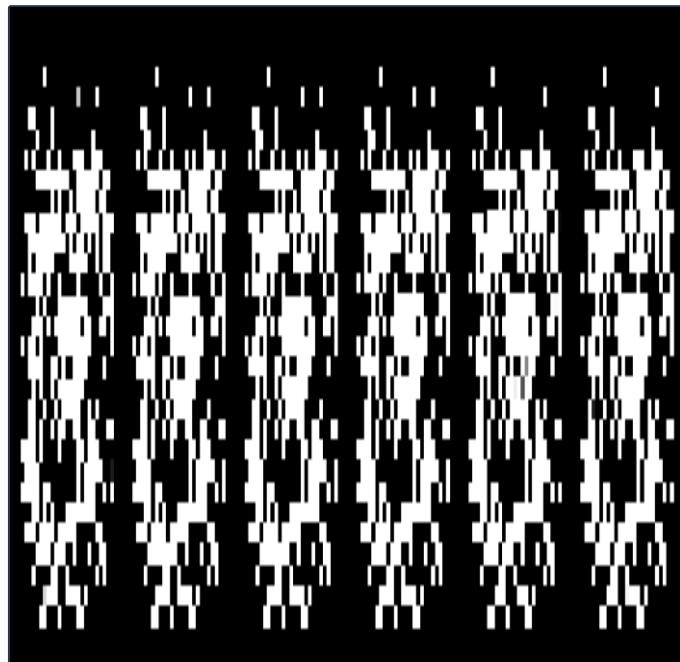


Generative Adversarial Network - MNIST

There is a common failure mode of GAN training termed *mode collapse*, that is when the generator learns to fool the discriminator producing only a certain subset of $P(x)$. Find hyperparameters that exhibit this failure mode and show generated examples.

Hyperparameters Leading to Mode Collapse:

- Batch Size: 16
- Generator Learning Rate: 0.002
- Discriminator Learning Rate: 0.0001
- Latent Dimension: 10



Simple Diffusion

Insert your simple diffusion visualizations here:

iteration 0

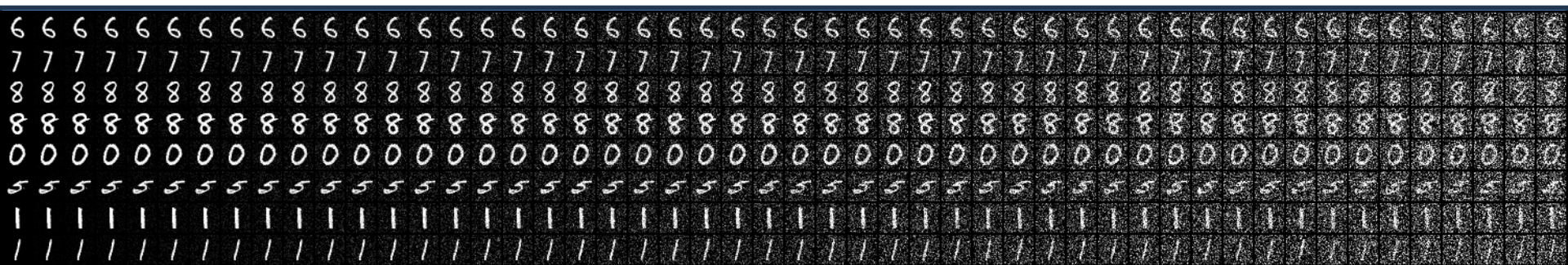


iteration 99



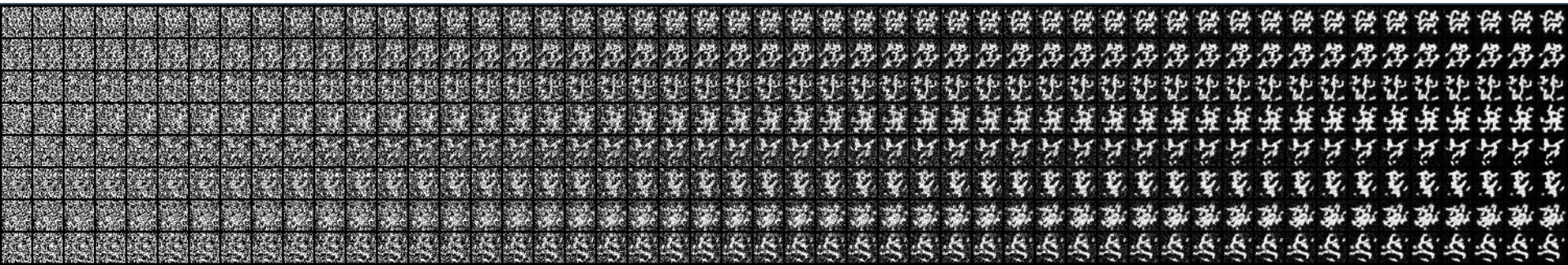
DDPM - MNIST

Insert your forward diffusion visual here (any epoch is fine):

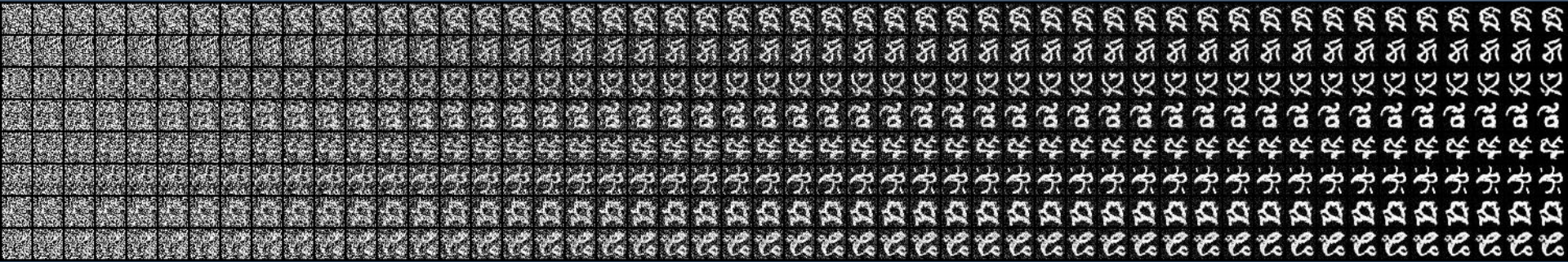


DDPM - MNIST

After tuning only noise schedule and timesteps:
Insert your reverse diffusion visual here (epoch 0):



Insert your reverse diffusion visual here (epoch 9):



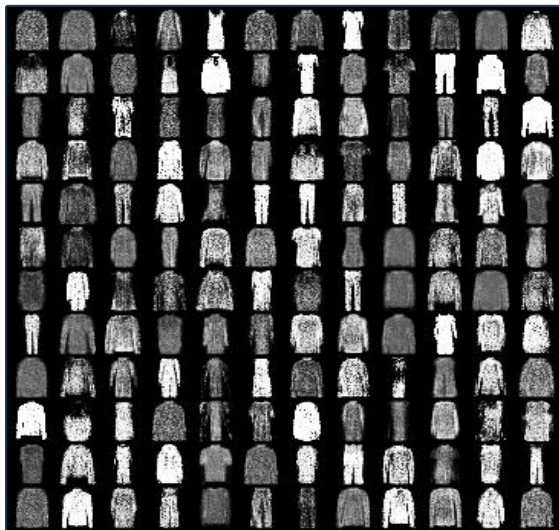
Tune each model (fashionMNIST) to achieve the best images and show your results:

VAE



FID: 103.66

GAN



FID: 138.370

Diffusion



FID: 42.857

Compare and contrast the results from the different models. Why might we see the results that we see? what role does the target dataset play?

In our experiments on FashionMNIST, the diffusion model achieved a significantly lower FID score compared to both the VAE and the GAN. The VAE, which minimizes a reconstruction loss and a KL divergence term to ensure a smooth, regularized latent space, produced samples that tended to be blurrier and less detailed, resulting in a higher FID. In contrast, while the GAN can produce sharper images, it is more susceptible to mode collapse and instability in training, which was reflected in its suboptimal FID score in our run. The diffusion model, however, leverages a gradual denoising process from pure noise, enabling it to capture finer details and generate more diverse, high-fidelity samples. This stepwise refinement allows the diffusion model to maintain better image quality and stability during training, ultimately yielding the lowest FID among the three approaches. These results underscore that for a complex dataset like FashionMNIST, where subtle texture and structural variations are critical, the iterative denoising mechanism of diffusion models is particularly effective.

FID is commonly used to benchmark generative models. Is it a good metric? Why or why not?

Fréchet Inception Distance (FID) has become a popular metric for comparing the performance of generative models, but it is not without limitations. On the one hand, FID measures how similar generated images are to real ones in a high-level feature space (extracted by an Inception network), which often correlates with perceptual quality and diversity. On the other hand, FID can be sensitive to factors like the specific feature extractor used, the number of samples, and potential overfitting to particular statistics. It also cannot capture all aspects of image quality or distribution coverage—there may be visually pleasing results that still yield a higher FID due to minor distribution mismatches, and conversely, a lower FID does not guarantee truly diverse outputs. Hence, while FID can serve as a useful benchmark, it is best interpreted alongside qualitative assessments and other metrics to gain a more complete picture of a model's generative performance.